

# AtlantikSolar

The first-ever crossing of the Atlantic Ocean using a solar-powered Unmanned Aerial Vehicle (UAV)



# Project Overview

*An Atlantic Crossing - on solar power alone!*

In May 1927, Charles Lindbergh put his life on the line to perform the first flight across the Atlantic in his gasoline-engine aircraft. In June 2015, in face of new pressing challenges such as climate change, the AtlantikSolar project will follow the footprints of the great aviator by demonstrating the first-ever fully solar-powered and autonomous crossing of the Atlantic Ocean.



The chosen flight route from Bell Island (Canada) to Lisbon (Portugal) is packed with challenges: 5000km of flight distance, 7 days in the harsh Atlantic environment with potentially strong winds or thunderstorms. Beginning June 21st 2015, a “swarm” of four AtlantikSolar Unmanned Aerial Vehicles (UAVs) - fueled by solar power alone - will be launched to face these challenges.

## Project Objectives

*Promoting civil Solar-UAVs and solar technology*

With the extreme showcase of an Atlantic Crossing, AtlantikSolar aims to promote the advantages of today’s solar-powered Unmanned Aerial Vehicles. Solar-UAVs such as the Autonomous Systems Lab’s (ASL) SkySailor have already demonstrated flights lasting more than one full day (24h), and the UAV behind AtlantikSolar is an advancement enabling even longer flight durations of up to 10 days. UAVs of this kind are a highly promising concept for multi-day or even multi-week search & rescue support, wildfire detection or meteorological surveys above otherwise inaccessible Oceans.

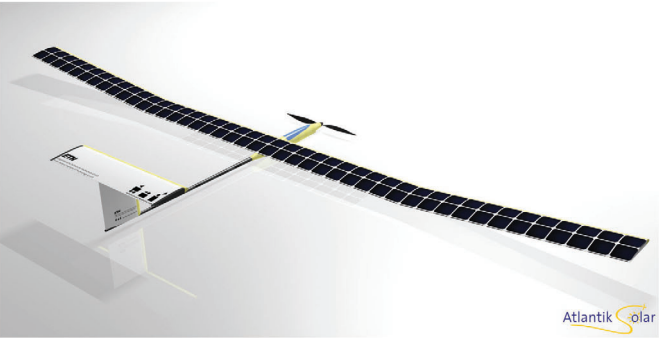
*A great challenge.  
A tremendous adventure.  
A unique chance to participate  
when aviation history is written.*

**AtlantikSolar.**

# The AtlantikSolar UAV

*A new generation of Solar UAV for a historic moment*

The UAV behind AtlantikSolar is a new generation of 5m-class solar-powered UAV, capable of performing flights of up to 10 days duration. Optimized for low power consumption and flight efficiency, it combines efficient solar-technology with state-of-the-art Li-Ion batteries. During the Atlantic crossing, the aircrafts will be steered by an autonomous control system developed at ETH Zurich and will transmit live images to the public.



| AtlantikSolar UAV – Technical Overview |                                                         |
|----------------------------------------|---------------------------------------------------------|
| Wingspan                               | 5.6m                                                    |
| Mass                                   | ~6.3kg                                                  |
| Structure                              | Lightweight carbon fibre & kevlar structure             |
| Power System                           | 1.4m² solar panels with Li-Ion batteries                |
| Flight Control                         | Autonomous control system (by ETHZ) with GPS-Navigation |
| Payload                                | Digital HD-camera, live-image transmission              |

The AtlantikSolar prototype aircraft performed its first test flight on November 14th 2012 near Pfäffikon, ZH. Thanks to the excellent flight characteristics, the general aerodynamical and structural concept could be verified. The successful first flight was the first major milestone of the project!



Landing of the AtlantikSolar prototype after its successful maiden flight on Nov. 14th 2012.

# The team

AtlantikSolar is headed by ETH Zurich's Autonomous Systems Lab, under supervision of Prof. R. Siegwart. Each core discipline is represented by one project partner. These current partners are:

**Autonomous Systems Lab, ETH Zurich**  
Project lead & overall aircraft design  
[www.asl.ethz.ch](http://www.asl.ethz.ch)



**Super Computing Systems**  
Electronics & satellite communication  
[www.scs.ch](http://www.scs.ch)

 **super computing systems**

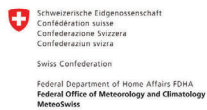
**Swiss Center for Electronics and Microtechnology**  
Solar Energy System  
[www.csem.ch](http://www.csem.ch)



**Skyguide**  
Flight regulations planning and support  
[www.skyguide.ch](http://www.skyguide.ch)



**MeteoSwiss**  
Meteorological planning and support  
[www.meteosuisse.ch](http://www.meteosuisse.ch)



Prof. Siegwart, ASL (left), Prof. Gunzinger, SCS (center), and part of the Atlantik-Solar team after the successful second prototype test flight.

## Project schedule

| Phase          | Milestone (and main tasks)                                                                                     | End date                         |
|----------------|----------------------------------------------------------------------------------------------------------------|----------------------------------|
| Phase 1a       | RC-controlled prototype first flight<br>Development of aircraft structure and basic electronics                | Nov. 14 <sup>th</sup> 2012       |
| Phase 1b       | First solar-powered flight<br>Development of solar panels and power management                                 | Sept. 5 <sup>th</sup> 2013       |
| Phase 1c       | Autopilot-controlled prototype first flight<br>Development and testing of advanced avionics and autopilot.     | March 26 <sup>th</sup> 2014      |
| Phase 2        | Multi-day (48h+) flight demonstration<br>Continuous flight testing as the final test for the Atlantic Crossing | July 1 <sup>st</sup> 2014        |
| <b>Phase 3</b> | <b>Atlantic Crossing</b>                                                                                       | <b>June 21<sup>st</sup> 2015</b> |
| Phase 3+       | Industrial applications                                                                                        | After 2015                       |

Up to the Atlantic crossing and later ultra-long duration flight applications, the project undergoes the major phases given above. Phase 1a&1b have already been completed, with the last achieved milestone being the successful demonstration of the first solar-powered flight of the AtlantikSolar prototype in early September 2013.

# Participate in a great adventure...

*... and help writing aviation history!*

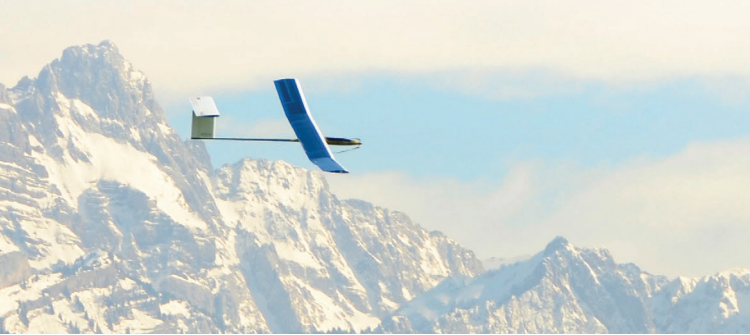
A solar-powered, autonomous crossing of the Atlantic is many things: A great challenge. A tremendous adventure. A unique chance to participate when aviation history is written. It is an excellent chance and ETH Zurich's inherent interest to reach out to the public; to convey the values and vision of the project alongside yours to your customers.

Your possible participation is organized into three main levels. As one of four Platinum sponsors - one per aircraft crossing the Atlantic - you are granted the exclusive option to advertize and "fly with your own aircraft" by choosing one aircraft name and design<sup>(6,7)</sup>. To bolster and profit of public interest, media events will be organized jointly for all sponsors before and during the Atlantic crossing. One of the highlights is a flying day in early 2015, with a live flying demo of the solar-powered UAV prototype and small exhibitions of related partner and major sponsor activities. In addition, all sponsors will be prominently represented in print and web products (e.g. <sup>(1,2,3)</sup>) and are encouraged to make maximum use of live flight reports and imagery. Join AtlantikSolar - and share the spirit of this historic undertaking with your customers!



## Solar Unmanned Aerial Vehicles at ETH Zurich

Solar powered unmanned civil aircraft have been the focus of intense research at ETH Zurich's Autonomous Systems Lab since 2002. Among them are SkySailor<sup>(4)</sup>, which in 2008 demonstrated the worldwide first solar-powered flight of more than 24-hours without the use of thermals, and SenseSoar<sup>(5)</sup>, an advanced scientific platform designed towards missions such as search & rescue support.



*An exceptional ambassador  
for Dedication, Innovation and Sustainability.  
An exceptional ambassador  
for You.*

**AtlantikSolar.**

Atlantik-Solar is a project administered by the Autonomous Systems Lab (ASL) of the Swiss Federal Institute of Technology Zurich (ETH Zurich).



- (1) [www.atlantiksolar.ethz.ch](http://www.atlantiksolar.ethz.ch)
- (2) [www.asl.ethz.ch](http://www.asl.ethz.ch)
- (3) [www.youtube.com/user/atlantiksolar](http://www.youtube.com/user/atlantiksolar)

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- (4) [www.asl.ethz.ch/research/asl/skysailor/](http://www.asl.ethz.ch/research/asl/skysailor/)
- (5) [www.sensesoar.ethz.ch](http://www.sensesoar.ethz.ch)
- (6) Name of aircraft as e.g. used in public print media or as watermark on live-imagery, to be defined jointly with the project team.
- (7) Design of aircraft rear section, to be defined jointly with the project team.